

COLORADO SPRINGS, CO, USA

WMO: 724660

Lat: 38.810N

Lon: 104.688W

Elev: 1884

StdP: 80.65

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93037

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.5	-13.7	-22.9	0.6	-7.0	-20.6	0.7	-3.4	14.5	1.6	12.7	2.0	2.8	150	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	32.8	14.8	31.5	14.7	30.1	14.6	17.6	25.8	16.9	25.3	16.4	24.8	5.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	13.9	19.2	14.7	13.1	18.9	13.9	12.5	18.7	56.9	25.7	54.7	25.2	52.8	24.8	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.2	9.5	DB	-21.4	35.3	3.3	1.2	-23.7	36.2	-25.7	36.9	-27.5	37.6	-29.9	38.5	(4)
(5)			WB	-21.8	19.2	3.2	0.8	-24.0	19.8	-25.9	20.2	-27.7	20.7	-30.0	21.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-0.2	0.4	4.7	8.2	13.4	19.3	22.1	20.8	17.0	10.1	4.3	-0.5	(6)	
	DBStd	9.39	5.70	5.91	5.49	5.13	4.52	3.69	2.63	2.56	4.17	5.17	5.52	5.78	(7)	
	HDD10.0	1452	318	269	176	92	21	1	0	0	6	62	182	325	(8)	
	HDD18.3	3341	576	501	422	303	161	33	4	8	70	257	422	583	(9)	
	CDD10.0	1459	1	2	13	38	126	279	375	336	216	64	9	1	(10)	
	CDD18.3	306	0	0	0	0	7	61	121	86	30	1	0	0	(11)	
	CDH23.3	3883	0	0	2	16	177	877	1376	919	464	52	1	0	(12)	
	CDH26.7	1423	0	0	0	1	36	343	592	320	128	4	0	0	(13)	
(14) Wind	WSAvg	4.2	3.9	4.3	4.6	5.2	4.8	4.5	3.9	3.8	4.0	4.1	3.9	3.9	(14)	
(15) Precipitation	PrecAvg	447	9	10	30	40	57	61	73	88	24	21	13	12	(15)	
	PrecMax	701	30	62	62	190	127	153	134	179	77	127	52	27	(16)	
	PrecMin	316	0	0	7	0	16	3	17	11	1	0	0	1	(17)	
	PrecStd	93	7	13	19	40	33	39	37	44	18	26	14	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	19.2	22.6	25.3	30.0	34.9	34.6	33.0	31.9	26.9	22.3	17.0	(19)	
		MCWB	5.4	5.8	7.6	9.3	12.4	13.7	15.7	15.4	14.5	11.3	7.7	5.2	(20)	
	2%	DB	14.7	15.9	20.0	22.9	27.4	32.3	33.1	31.4	29.7	24.9	19.5	14.4	(21)	
		MCWB	4.0	4.6	6.4	8.5	11.3	13.6	15.2	15.0	13.4	10.2	6.6	4.0	(22)	
	5%	DB	12.0	13.5	17.7	20.9	25.3	30.5	31.9	30.0	28.1	22.6	17.1	12.0	(23)	
		MCWB	2.9	3.5	5.6	7.6	10.7	13.4	15.3	15.2	13.1	9.5	5.7	2.9	(24)	
	10%	DB	9.6	10.7	15.1	18.6	23.0	28.5	30.3	28.5	26.3	20.0	14.3	9.4	(25)	
		MCWB	1.6	2.1	4.6	6.9	10.4	13.3	15.3	15.2	12.7	8.5	4.6	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	6.5	8.3	10.7	14.7	17.4	18.6	18.4	16.8	13.0	8.4	5.7	(27)	
		MCDB	16.4	17.6	20.0	20.5	23.5	25.9	25.9	26.0	25.8	21.0	19.6	15.9	(28)	
	2%	WB	4.2	5.0	7.1	9.5	13.4	16.2	17.8	17.5	15.5	11.7	7.1	4.3	(29)	
		MCDB	13.7	15.0	18.2	19.5	22.1	24.9	25.8	25.7	24.3	20.8	18.3	13.5	(30)	
	5%	WB	3.0	3.8	6.1	8.5	12.3	15.4	17.2	16.9	14.7	10.5	6.0	3.0	(31)	
		MCDB	11.5	12.9	16.4	18.6	21.1	24.9	25.5	25.0	23.5	19.9	16.2	11.5	(32)	
	10%	WB	1.8	2.4	5.0	7.6	11.5	14.8	16.6	16.3	13.9	9.3	4.9	1.7	(33)	
		MCDB	9.0	10.3	14.3	17.3	20.1	24.4	25.1	24.4	22.7	18.5	13.7	9.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	13.5	14.2	13.6	13.7	14.8	14.4	13.7	14.4	14.2	13.8	13.2	(35)	
		MCDBR	17.2	17.6	17.8	18.0	17.7	18.1	17.1	16.2	17.3	18.1	18.1	16.9	(36)	
	5% WB	MCWBR	9.3	9.0	8.2	7.8	6.5	5.8	5.1	5.0	5.9	7.7	8.7	9.1	(37)	
		MCWBR	16.7	17.1	16.9	16.6	14.9	15.0	13.8	13.3	14.5	15.7	16.9	16.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.232	0.236	0.257	0.283	0.302	0.315	0.343	0.331	0.296	0.262	0.241	0.234	(40)	
		taud	2.629	2.648	2.584	2.520	2.528	2.510	2.479	2.503	2.574	2.652	2.631	2.620	(41)	
	Ebn at Noon		983	1019	1016	997	976	959	928	935	957	971	962	956	(42)	
		Edh at Noon	66	75	89	102	103	105	108	103	90	74	66	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.72	3.66	4.91	5.95	6.58	7.33	6.93	6.30	5.44	4.08	2.99	2.40	(44)	
	RadStd		0.14	0.23	0.36	0.37	0.49	0.33	0.29	0.21	0.28	0.24	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.56	N/A	N/A	+0.58	N/A	N/A	N/A	-140	+123	+61	(46)
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+67	+22	(47)

Nomenclature: See separate page